

Coding & Solution

Date	5 July 2026
Team ID	–
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section showcases the quality and correctness of our implemented code and the overall solution. Well-structured code, a clear modular working functionality, clean code practices and alignment with project requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solutions should address the core problem statement and requirement for key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., **GitHub**) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/your-team/voice-concept-analyser
Programming Language(s) Used	Python
Framework(s) Used	FastAPI, React.js, Tailwind CSS
Key Features Implemented	• Voice input (upload/record) and transcription • NLP-based concept matching and similarity scoring • Detailed feedback and suggestions • Progress tracking and report generation
Pending / Incomplete Features	Multi-language support, User roles (Teacher/Student/Admin)
Setup / Run Instructions	1. Clone the repository 2. Create virtual environment: <code>python -m venv venv</code> 3. Install dependencies: <code>pip install -r requirements.txt</code> 4. Run backend: <code>uvicorn app.main:app --reload</code> 5. Run frontend: <code>cd frontend && npm install && npm run dev</code>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for all critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows clean code principles with modular design and separation of concerns. Comprehensive comments are added in complex logic areas. Future enhancements like multi-language support and advanced analytics are planned in upcoming sprints.